DET-007 | Base64 Obfuscated Payload Detection

ID: DET-007
MITRE ID: T1027
Data Source: DeviceProcessEvents

Severity: High
Tactic:Defense Evasion
Category: Endpoint

ADX MAPPING LOGIC:
Mapping: EpisodeId (string) = 'CommandLine containing Base64'. We use matches regex and strlen() to simulate detection of long Base64 strings in command-line arguments — core ADX regex operators.

OBJECTIVE: Practice matches regex and strlen() — the operators used to detect Base64 obfuscation in command lines. Real Sentinel: any process command line containing a Base64 string >100 chars.

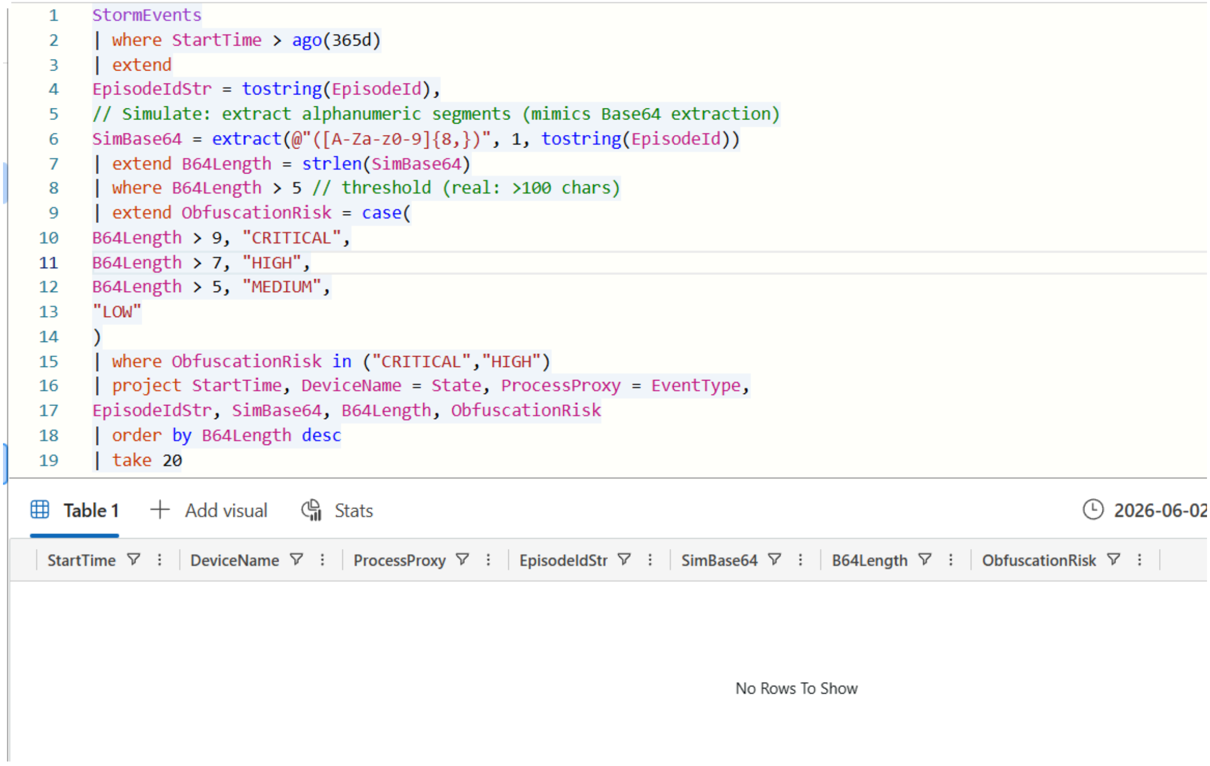
KQL QUERY (RUNNABLE ON ADX FREE TIER)

```
StormEvents
| where StartTime > ago(365d)
| extend
EpisodeIdStr = tostring(EpisodeId),
// Simulate: extract alphanumeric segments (mimics Base64 extraction)
SimBase64 = extract(@"([A-Za-z0-9]{8,})", 1, tostring(EpisodeId))
| extend B64Length = strlen(SimBase64)
| where B64Length > 5 // threshold (real: >100 chars)
| extend ObfuscationRisk = case(
B64Length > 9, "CRITICAL",
B64Length > 7, "HIGH",
B64Length > 5, "MEDIUM",
"LOW"
)
| where ObfuscationRisk in ("CRITICAL","HIGH")
| project StartTime,DeviceName = State,
ProcessProxy = EventType,
EpisodeIdStr,
SimBase64,B64Length,ObfuscationRisk
| order by B64Length desc
| take 20
```

INVESTIGATION STEPS
1. Extract and decode the Base64 string in CyberChef.
2. Check if decoded content contains URLs, shellcode or scripts.
3. Trace the parent process chain back to origin.
4. Review similar events on other devices (threat hunt).

RESPONSE ACTIONS
- Isolate devices if decoded content is malicious.
- Extract IOCs from decoded payload (IPs, domains, hashes).
- Add IOCs to threat intelligence feeds.
- Enable PowerShell ScriptBlock logging if not already active.

Output from Above KQL Query:



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| extend ObfuscationRisk = case(
B64Length > 9, "CRITICAL",
B64Length > 7, "HIGH",
B64Length > 5, "MEDIUM",
"LOW"
)
| where ObfuscationRisk in ("CRITICAL","HIGH")
| project StartTime, DeviceName = State, ProcessProxy = EventType,
EpisodeIdStr, SimBase64, B64Length, ObfuscationRisk
| order by B64Length desc
| take 20
```


KQL Detection Query in Sentinel:

```
DeviceProcessEvents
| where TimeGenerated > ago(1h)
| where ProcessCommandLine matches regex @"[A-Za-z0-9+/]{100,}={0,2}"
| where ProcessCommandLine has_any (
"powershell","cmd","wscript",
"cscript","mshta","rundll32"
)
| extend B64Length = strlen(
extract(@"([A-Za-z0-9+/]{100,}={0,2})", 1, ProcessCommandLine)
)
| where B64Length > 100
| project
TimeGenerated, DeviceName, AccountName,
FileName, ProcessCommandLine, B64Length,
InitiatingProcessFileName
| order by B64Length desc
```